3 CONTINUOUS DISTRIBUTIONS

3.1 Objectives

By the end of the session, you should be able to :

i) Describe Uniform distribution and solve related questions

ii) Use the normal tables

iii) Apply the normal distribution to real life situations

3.2 UNIFORM OR RECTANGULAR DISTRIBUTION

A Random variable X is defined to have a rectangular distribution if the continuous density function of X is given by

$$f(x) = \frac{1}{b-a}, \qquad a < x < b$$
$$= 0, \ elsewhere$$

The Mean, Variance and Moment generating function of the Uniform or Rectangular Distribution

$$Mean = E(x) = \int_{x=a}^{b} x \cdot f(x)\,dx$$

$$= \int_{x=a}^{b} x \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_{x=a}^{b} x\,dx$$

$$= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_{x=a}^{b}$$

$$= \frac{1}{2(b-a)}\left[x^2\right]_{x=a}^{b} = \frac{1}{2(b-a)}\left[b^2 - a^2\right]$$

$$= \frac{1}{2(b-a)}\left[(b-a)(b+a)\right] = \frac{b+a}{2}$$

$$\therefore \ Mean = \frac{a+b}{2}$$

$$Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$

Now we want to evaluate E (x^2).

$$E(x^2) = \int_{x=a}^{b} x^2 \cdot f(x)\,dx$$

$$= \int_{x=a}^{b} x^2 \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_{x=a}^{b} x^2\,dx$$

$$= \frac{1}{b-a}\left[\frac{x^3}{3}\right]_{x=a}^{b}$$

$$= \frac{1}{3(b-a)}\left[x^3\right]_{x=a}^{b} = \frac{1}{3(b-a)}\left[b^3 - a^3\right]$$

$$= \frac{1}{3(b-a)}\left[(b-a)(b^2+ab+a^2)\right] = \frac{(b^2+ab+a^2)}{3}$$

$$\therefore\ E(x^2) = \frac{b^2+ab+a^2}{3}$$

$$\therefore\ Var(x) = \sigma^2 = E(x^2) - \left[E(x)\right]^2$$

$$= \frac{(b^2+ab+a^2)}{3} - \left[\frac{a+b}{2}\right]^2$$

$$= \frac{(b^2+ab+a^2)}{3} - \frac{(a+b)^2}{4}$$

$$= \frac{4(b^2+ab+a^2) - 3(a^2+2ab+b^2)}{12}$$

$$\Rightarrow\ Var(x) = \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}$$

We want to evaluate the m.g.f.

$$M_X(t) = E(e^{tx})$$

$$= \int_{x=a}^{b} e^{tx} \cdot f(x)dx$$

$$= \int_{x=a}^{b} e^{tx} \cdot \frac{1}{b-a}dx = \frac{1}{b-a}\int_{x=a}^{b} e^{tx}dx$$

$$= \frac{1}{b-a}\left[\frac{e^{tx}}{t}\right]_{x=a}^{b}$$

$$= \frac{1}{t(b-a)}\left[e^{tb} - e^{ta}\right]$$

$$\Rightarrow M_X(t) = \frac{1}{t(b-a)}\left[e^{tb} - e^{ta}\right]$$

3.3 NORMAL DISTRIBUTION

The continuous distribution having the density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x, \mu < \infty \quad \text{.....(1)}$$

$$= 0, \textit{ otherwise}$$

is called the **normal distribution or Gauss distribution**.

A random variable having this distribution is said to be **normal or normally distributed**. This distribution is very important, because many random variables of practical interest are normal or approximately normal or can be transformed into normal random variables in a relatively simple fashion. Furthermore, the normal distribution is a useful approximation of more complicated distributions. It also appears in the mathematical proofs of various statistical tests.

In Eq. (1), □ is the mean and □ is the standard deviation of the distribution.

The curve of f (x) is called the bell-shaped curve. It is symmetric with respect to □ .

The figure above shows f (x) for □ = 0. For □ >0 (□ <0) the curves have the same shape, but are shifted $|\mu|$ units to the right (left). The smaller σ^2 is, the higher is the peak at x = 0 in figure (1) above and the steeper are the descents on both sides. This agrees with the meaning of variance.
From Eq. (1) we see that the normal distribution has the cumulative distribution function

$$\Pr(X \le x) = F(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{w-\mu}{\sigma}\right)^2} dw, \quad \text{.....(2)}$$

From this we obtain

$$\Pr(a \le X \le b) = F(b) - F(a) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{a}^{b} e^{-\frac{1}{2}\left(\frac{w-\mu}{\sigma}\right)^2} dw, \quad \text{.....(3)}$$

The integral in (2) cannot be evaluated by elementary methods, but can be represented in terms of the integral

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{t^2}{2}} dt \quad \text{.....(4)}$$

which is the distribution function of the normal distribution with mean = 0 and variance = 1 and has been tabulated (see mathematical tables).
Infact, if we set

$t = \frac{w-\mu}{\sigma}$, then $\frac{dt}{dw} = \frac{1}{\sigma} \quad \Rightarrow dw = \sigma\, dt$, and we have to integrate from $-\infty$ *to* $z = \frac{x-\mu}{\sigma}$.
From Eq. (2) we thus obtain

$$F(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{t^2}{2}} \sigma\, dt$$

σ drops out, and the expression on the right equal (4), where $z = \frac{x-\mu}{\sigma}$. That is

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right) \quad \text{.....(5)}$$

The cumulative distribution of z is known as the standard normal distribution.
From this important formula and Eq. (3) we obtain another important formula

$$\Pr(a \le X \le b) = F(b) - F(a) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

- Let X be binomial with parameters n and p. For large n, X is approximately normal with mean, $\mu = np$ and variance $\sigma^2 = npq = np(1-p)$.
  For practical purposes, use normal approximation to binomial distribution for either p > 0.5 and np > 5 **or** p > 0.5 and n (1-p) > 5.
- Let X be Poisson with parameter μ. Then for large values of n, X is approximately normal with mean μ and variance μ.

The Moment generating function, Mean, and Variance

We first work out the m.g.f. which will in turn be used to obtain the mean and variance expressions.

By definition;

$$M.g.f. = M_X(t) = E(e^{tx})$$
$$= E\left[e^{tx+\mu t-\mu t}\right]$$
$$= E\left[e^{tx-\mu t}\cdot e^{\mu t}\right]$$
$$= e^{\mu t}E\left[e^{tx-\mu t}\right] = e^{\mu t}E\left[e^{t(x-\mu)}\right]$$
$$= \int_{x=-\infty}^{\infty} e^{\mu t}e^{t(x-\mu)}\, f(x)\,dx$$
$$= \int_{x=-\infty}^{\infty} e^{\mu t}e^{t(x-\mu)}\frac{1}{\sqrt{2\pi\sigma^2}}e^{\frac{-(x-\mu)^2}{2\sigma^2}}\,dx$$
$$= e^{\mu t}\int_{x=-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}\,e^{\frac{-1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2t(x-\mu)\right]}dx$$

$$= e^{\mu t}\int_{x=-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}\,e^{\frac{-1}{2\sigma^2}\left[(x-\mu)^2-2t\sigma^2(x-\mu)\right]}dx$$

Consider from:

$$(x-\mu)^2 - 2t\sigma^2(x-\mu) = (x-\mu)^2 - 2t\sigma^2(x-\mu) + \sigma^4t^2 - \sigma^4t^2$$
$$= \left[(x-\mu)-\sigma^2t\right]^2 - \sigma^4t^2$$

Then;

$$M_X(t) = e^{\mu t} \int_{x=-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{\frac{-1}{2\sigma^2}\left\{\left[(x-\mu)-\sigma^2 t\right]^2 - \sigma^4 t^2\right\}}\, dx$$

$$= e^{\mu t} \int_{x=-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{\frac{-1}{2\sigma^2}\left\{\left[(x-\mu)-\sigma^2 t\right]^2\right\}} \cdot e^{\frac{\sigma^4 t^2}{2\sigma^2}}\, dx$$

$$= e^{\mu t} \cdot e^{\frac{\sigma^2 t^2}{2}} \int_{x=-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{\frac{-1}{2\sigma^2}\left\{\left[(x-\mu)-\sigma^2 t\right]^2\right\}}\, dx$$

$$= e^{\mu t + \frac{\sigma^2 t^2}{2}} \int_{x=-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{\frac{-1}{2\sigma^2}\left\{\left[(x-\mu)-\sigma^2 t\right]^2\right\}}\, dx$$

The integral $\int_{x=-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{\frac{-1}{2\sigma^2}\left\{\left[(x-\mu)-\sigma^2 t\right]^2\right\}}\, dx$ is 1 since it is the area under a normal distribution with mean $\mu + \sigma^2 t$ and variance = σ^2.

Hence $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$

Next, we know that

$Mean = M_X'(0)$

$$\therefore\ M_X'(t) = \frac{d}{dt}\left[e^{\mu t + \frac{\sigma^2 t^2}{2}}\right] = (\mu + \sigma^2 t) e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

At t = 0; we have

$M_X'(0) = (\mu + \sigma^2 \cdot 0) e^0 = \mu$

$\Rightarrow\ Mean = \mu$

Next we work out Var (x)

We know that

$$Var(x) = \sigma^2 = E(x^2) - [E(x)]^2$$
$$= M_X''(0) - [M_X'(0)]^2$$

$$M_X'(t) = (\mu + \sigma^2 t)e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

Using product rule to differentiate the above expression, we get

Let

$$U = (\mu + \sigma^2 t) \Rightarrow \frac{dU}{dt} = \sigma^2$$

$$V = e^{\mu t + \frac{\sigma^2 t^2}{2}} \Rightarrow \frac{dV}{dt} = (\mu + \sigma^2 t)e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$\therefore \; M_X''(t) = V\frac{dU}{dt} + U\frac{dV}{dt}$$
$$= e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot \sigma^2 + (\mu + \sigma^2 t) \cdot (\mu + \sigma^2 t)e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$At \;\; t = 0$$

$$M_X''(0) = \; e^0 \cdot \sigma^2 + (\mu + \sigma^2 \cdot 0) \cdot (\mu + \sigma^2 \cdot 0)e^0$$
$$= \mu^2 + \sigma^2$$

But

$$Var(x) = M_X''(0) - [M_X'(0)]^2$$
$$= \mu^2 + \sigma^2 - (\mu)^2$$
$$= \sigma^2$$

Hence

$Var(x) = \sigma^2$ which is the variance of the normal variable X.

Standard Normal Variable, Z

Let $Z = \frac{X - \mu}{\sigma}$

We say

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}}$$

$$E(Z) = E\left[\frac{X-\mu}{\sigma}\right] = \frac{E(X) - E(\mu)}{\sigma}$$

But $E(X) = \mu$

$$\therefore\ E(Z) = \frac{\mu - \mu}{\sigma} = 0$$

$$Var(Z) = E(Z^2) - [E(Z)]^2$$
$$= E(Z^2) \quad Since\ E(Z) = 0$$

But

$$E[Z^2] = E\left[\left(\frac{X-\mu}{\sigma}\right)\cdot\left(\frac{X-\mu}{\sigma}\right)\right]$$

$$= \frac{1}{\sigma^2} E[(X-\mu)^2] = \frac{1}{\sigma^2}[E(X^2 - 2X\mu + \mu^2)]$$

$$= \frac{1}{\sigma^2}[E(X^2) - E(2X\mu) + E(\mu^2)]$$

$$= \frac{1}{\sigma^2}[E(X^2) - 2\mu E(X) + \mu^2] = \frac{1}{\sigma^2}[E(X^2) - 2\mu^2 + \mu^2]$$

$$= \frac{1}{\sigma^2}[E(X^2) - \mu^2] = \frac{1}{\sigma^2}[E(X^2) - [E(X)]^2]$$

$$= \frac{1}{\sigma^2}\cdot\sigma^2 = 1$$

$\therefore\ Var(Z) = 1$

$\Rightarrow$ Z is normally distributed with mean = 0 and variance = 1.

i.e. $Z \sim N(0, 1)$

Example 1

Determine the probabilities

(a) $\Pr(X \le 2.44)$, (b) $\Pr(X \le -1.16)$, (c) $\Pr(X \ge 1)$

(d) $\Pr(2 \le X \le 10)$ where X is assumed to be normal with mean 0 and variance 1.

Solution

Since $\mu = 0$ *and* $\sigma^2 = 1$, we may obtain the desired values directly from the mathematical tables;

(a) 0.9927, (b) 0.1230, (c) $\Pr(X \ge 1) = 1 - \Pr(X < 1) = 1 - 0.8413 = 0.1587$

(d) $\Pr(2 \le X \le 10) = \Phi(10) - \Phi(2) = 1 - 0.9772 = 0.0228$

Example 2

Let X be normal with mean 0 and variance 1. Determine the constant c such that:

(a) $\Pr(X \ge c) = 10\%$, (b) $\Pr(X \le c) = 5\%$, (c) $\Pr(0 \le X \le c) = 45\%$

(d) $\Pr(-c \le X \le c) = 99\%$

Solution

Since $\mu = 0$ *and* $\sigma^2 = 1$, we have;

(a)

$$\Pr(X \ge c) = 1 - \Pr(X \le c) = 1 - \Phi\left(\frac{c-0}{1}\right) = 0.1$$

$$\Rightarrow \Phi(c) = 1 - 0.1 = 0.9 \quad \Rightarrow c = \Phi^{-1}(0.9) = 1.282$$

(b)

$$\Pr(X \le c) = \Phi\left(\frac{c-0}{1}\right) = 0.05$$

$$\Rightarrow \Phi(c) = 0.05 \quad \Rightarrow c = \Phi^{-1}(0.05) = -\Phi^{-1}(1 - 0.05) = -\Phi^{-1}(0.95) = -1.645$$

(c)

$$\Pr(0 \le X \le c) = \Phi\left(\frac{c-0}{1}\right) - \Phi\left(\frac{0-0}{1}\right) = \Phi(c) - \Phi(0) = 0.45$$

$$\Rightarrow \Phi(c) - 0.5 = 0.45 \quad \Rightarrow \Phi(c) = 0.5 + 0.45 = 0.95 \quad \Rightarrow c = \Phi^{-1}(0.95) = 1.645$$

(d)

$$\Pr(-c \le X \le c) = \Phi\left(\frac{c-0}{1}\right) - \Phi\left(\frac{-c-0}{1}\right) = \Phi(c) - \Phi(-c) = 0.99$$

$$\Rightarrow \Phi(c) - \left[1 - \Phi(c)\right] = 0.99 \quad \Rightarrow 2\Phi(c) = 1 + 0.99 = 1.99 \quad \Rightarrow \Phi(c) = 0.995$$

$$\Rightarrow c = \Phi^{-1}(0.995) = 2.576$$

Example 3

Let X be normal with mean = -2 and variance = 0.25. Determine the constant c such that:

(a) $\Pr(X \ge c) = 0.2$, (b) $\Pr(-c \le X \le -1) = 0.5$,

(c) $\Pr(-2-c \le X \le -2+c) = 0.9$ (d) $\Pr(-2-c \le X \le -2+c) = 99.6\%$

Solution

Since $\mu = -2$ *and* $\sigma^2 = 0.25 \Rightarrow \sigma = 0.5$, we have;

(a)

$$\Pr(X \ge c) = 1 - \Pr(X \le c) = 1 - \Phi\left(\frac{c+2}{0.5}\right) = 0.2$$

$$\Rightarrow \Phi\left(\frac{c+2}{0.5}\right) = 1 - 0.2 = 0.8 \quad \Rightarrow \frac{c+2}{0.5} = \Phi^{-1}(0.8) = 0.842 \Rightarrow c = -1.579$$

(b)

$$\Pr(-c \le X \le -1) = \Phi\left(\frac{-1+2}{0.5}\right) - \Phi\left(\frac{-c+2}{0.5}\right) = 0.5$$

$$\Rightarrow \Phi(2) - \Phi\left(\frac{-c+2}{0.5}\right) = 0.5 \quad \Rightarrow 0.9772 - \Phi\left(\frac{-c+2}{0.5}\right) = 0.5 \Rightarrow \Phi\left(\frac{-c+2}{0.5}\right) = 0.9772 - 0.5$$

$$\Rightarrow \Phi\left(\frac{-c+2}{0.5}\right) = 0.4772 \Rightarrow \frac{-c+2}{0.5} = \Phi^{-1}(0.4772) = -\Phi^{-1}(1 - 0.4772) = -\Phi^{-1}(0.5228)$$

$$\Rightarrow \frac{-c+2}{0.5} = -0.057 \Rightarrow c = 2.0285$$

(c)

$$\Pr(-2-c \le X \le -2+c) = \Phi\left(\frac{-2+c+2}{0.5}\right) - \Phi\left(\frac{-2-c+2}{0.5}\right) = 0.9$$

$$\Rightarrow \Phi(2c) - \Phi(-2c) = 0.9 \quad \Rightarrow \Phi(2c) - [1-\Phi(2c)] = 0.9 \Rightarrow 2\Phi(2c) = 1+0.9 = 1.9$$

$$\Rightarrow \Phi(2c) = 0.95 \Rightarrow 2c = \Phi^{-1}(0.95) = 1.645$$

$$\Rightarrow c = 0.823$$

(d)

$$\Pr(-2-c \le X \le -2+c) = \Phi\left(\frac{-2+c+2}{0.5}\right) - \Phi\left(\frac{-2-c+2}{0.5}\right) = 0.996$$

$$\Rightarrow \Phi(2c) - \Phi(-2c) = 0.996 \;\Rightarrow \Phi(2c) - [1-\Phi(2c)] = 0.996 \Rightarrow 2\Phi(2c) = 1+0.996 = 1.996$$

$$\Rightarrow \;\Phi(2c) = 0.998 \Rightarrow 2c = \Phi^{-1}(0.998) = 2.878$$

$$\Rightarrow c = 1.439$$

Example 4

An industrial process mass produces items which are normally distributed. 11.55% of them weigh over 20 kg and 5.89% weigh under 10 kg. Calculate the mean weight and standard deviation for this distribution.

Solution

(a)

$$\Pr(X > 20) = 1 - \Pr(X \le 20) = 1 - \Phi\left(\frac{20-\mu}{\sigma}\right) = 0.1155$$

$$\Rightarrow \Phi\left(\frac{20-\mu}{\sigma}\right) = 1 - 0.1155 = 0.8845 \;\Rightarrow \frac{20-\mu}{\sigma} = \Phi^{-1}(0.8845) = 1.2$$

$$\Rightarrow 20 - \mu = 1.2\sigma$$

$$\Rightarrow \mu + 1.2\sigma = 20 \ldots\ldots\ldots\

$\mu = 15.652 \quad and \quad \sigma = 3.623$

EXERCISE

1. Let X be normal with mean = 80 and variance = 9. Find:
   (a) $\Pr(X > 83)$ **[0.1587]** (b) $\Pr(X < 81)$ **[0.6306]**
   (c) $\Pr(X < 80)$ **[0.5]** (c) $\Pr(78 \le X < 82)$ **[0.4950]**
2. Let X be normal with mean = 14 and variance = 4. Determine the constant c such that:
   (a) $\Pr(X \le c) = 95\%$ **[17.29]** (b) $\Pr(X \le c) = 5\%$ **[-17.29]**
   (c) $\Pr(-c \le X \le c) = 99\%$ **[19.152]**
3. If the lifetime X of a certain kind of automobile battery is normally distributed with a mean of 4 years and a standard deviation of 1 year, and the manufacturer wishes to guarantee the battery for 3 years, what percentage of the batteries will he have to replace under the guarantee? **[16%]**
4. A manufacturer produces airmail envelopes whose weight is normal with mean $\mu = 1.950 \ \ grams$ and standard deviation $\sigma = 0.025 \ grams$.
   The envelopes are sold in lots of 1000. How many envelopes in a lot will be heavier than 2 grams? **[About 22]**
5. Specifications for a certain job call for bolts with a diameter of $0.280 \pm 0.002 \ cm$.
   If the diameters of the bolts made by some manufacturer are normally distributed with mean $\mu = 0.279$ and standard deviation $\sigma = 0.001$, what percentage of these bolts will meet specifications? **[84%]**
6. The lengths of bolts produced in a certain factory may be taken to be normally distributed. The bolts are checked on two "go-no-go" gauges so that those shorter than 2.983 in. or longer than 3.021 in. are rejected as "too-short" or "too-long" respectively.
   (i) A random sample of N = 300 bolts is checked. If they have mean length 3.007 in. and standard deviation 0.011 in., what values would you expect for n_1, the number of "too-short" bolts and n_2, the number of "too-long" bolts? **[n_1 = 4.4, n_2 = 30.6]**
   (ii) A sample of N = 600 bolts from another factory is also checked. If for this sample we find n_1 = 20 and n_2 = 15, find estimates for the mean and standard deviation of the lengths of these bolts. **[= 3.001, = 0.01]**